

Failure 3 — the cost wall

Better rerankers exist. You can't afford to run them.

LATENCY TAX

40x

slower per query

cross-encoder → LLM reranker

Production search has a second-scale latency budget, not a minute-scale one.

You can afford the LLM reranker on 10–20 candidates. Not on 1,000.

Better rerankers don't fix bad retrievers — they make the gap more expensive to ignore.

TIME TO RANK 1,000 DOCUMENTS

cross-encoder reranker

~3 s

LLM reranker (7B param)



1–2 s

production budget